COURSE (Subject) BOOK

Foundations of Mathematics I

College Name	Education
Department Name	Mathematics
Academic Year	2026 – 2027
Lecturer Name	Dr Hogir Mohammed Yaseen and Shaymaa Maghdid Mohammed
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Lecturers' Profile Link	B.Sc. in Mathematics (2007, SUE), M.Sc. in Algebra (2010, SUE), PhD in representation of Lie algebras (University of Leicester, 2018)
Office Hours	Sunday to Thursday 10 AM–2 PM Contact via email / Tel: 07504154982
Year of Study	First stage
Level of Course	Undergraduate
1. Course Information:	
Course Title	Foundations of Mathematics I
Course Code	S070402
Credit Hours (ECTS)	6 ECTS
Semester	1st (Fall)
Prerequisites	No

Theory	4 hours
Practical (if applicable)	2 hours (Discussion)

2. Introduction:

In the first chapter, we study basic mathematical logic that consists of statements, truth values, and logical operations (negation, conjunction, disjunction, conditional, biconditional) in terms of logical truth tables. On the other hand, we study existential and universal quantifiers. Furthermore, there are some laws of algebra (Idempotent, Associative, Commutative, Distributive, Complement, and Identity) to prove some theorems (or logical equivalent statements). In the second chapter, students will review some operations of sets (union of sets, intersection of sets, difference and symmetric difference of sets, complement of set, ordered pair, ordered n-tuple, equality of ordered n-tuples, Cartesian product of sets). Additionally, they will learn new ideas about sets and theorems in set theory. The third chapter is about relations. In spite of revising the concept of relation and their compositions, students will study new definitions and theorems. Moreover, we study more properties of relations and their elements such as least element, greatest element, minimal elements, bounded below, bounded above, infimum, supremum, totally ordered, and well-ordered relations. Furthermore, we illustrate these definitions by solving many examples.

3. Aim of the Course:

Foundations of mathematics is the study of basic mathematical concepts (Mathematical logic, set theory, Relation, function, Construction of numbers, Group, Ring, Field, Cardinality) and how they form hierarchies of more complex structures and concepts, especially the fundamentally important structures that form the language of mathematics. The study aims to deepen understanding of these fundamental structures, enhance logical reasoning skills through formal logic and proofs, and explore foundational mathematical structures and their properties.

4. Objectives*:*

Upon completion of this course, students should be able to:

i. Develop an appreciation for the rigor and precision required in mathematical arguments and proofs.
ii. Learn about axiomatic systems, which define mathematical theories.
iii. Demonstrate strong problem-solving abilities and a solid foundation in mathematical thinking and reasoning.
iv. Prepare for advanced mathematical topics and understand how foundational mathematics connects with other fields like physics and computer science.

5. Course Learning Outcomes:

Category	CLOs	course Learning Outcome
Knowledge	CLO1	Demonstrate understanding of fundamental mathematical logic, truth tables, set operations,

		ordered pairs, and relation properties.
Intellectual Skills: Application of Intellectual Skills	CLO2	Apply mathematical logic, set theory, and relation concepts to analyze and solve mathematical problems.
Intellectual Skills: Analysis and Evaluation of Intellectual Skills	CLO3	Analyze mathematical statements and structures using appropriate proof methods, logical reasoning, and foundational concepts.
Practical Skills	CLO4	Apply set theory concepts, Venn diagrams, set identities, Cartesian products, and relation compositions to model and solve mathematical problems.
Communication skills	CLO5	Communicate mathematical ideas, arguments, and proofs effectively in written and oral forms using appropriate mathematical language.
Professional/Organizational skills	CLO6	Demonstrate independent learning, critical thinking, problem-solving, and effective participation in mathematical discussions and activities.

6. Reading and Learning Resources:

Essential Textbooks	1. H. Behnke, F. Bachmann, and H. Fladt. Fundamentals of Mathematics, 1974. 2. Alan G. Hamilton. Numbers, Sets and Axioms: The Apparatus of Mathematics, Cambridge University Press, 1982. 3. Elliott Mendelson. Number Systems and the Foundations of Analysis, 1973. 4. Ian Stewart and David Tall. The Foundations of Mathematics, OUP Oxford, 2015. 5. Raymond L. Wilder et al. Introduction to the Foundations of Mathematics, Courier Corporation, 2012. 6. أسس الرياضيات جزء الاول والثانى
Recommended Reading	PowerPoint presentation for course outline, titles, definitions, discussions, and conclusions. Blackboard for solving and explaining examples. (2 hours discussion + 4 hours theoretical).

Online Resources	https://drive.google.com/drive/folders/1Sbw_D_ZzCfdEsfF8PomwGP19krjkI427?usp=drive_link

7. Main Topics of the Subject (course):

Week	Topic
1	Registration
2	Introduction to Foundations of Mathematics
3	Mathematical Logic, Statements and their Types
4	Tautology, Contradiction, Logical Equivalent
5	Open Sentence, Quantifiers (Existential, Universal)
6	Set Theory (Algebra of Sets)
7	Concept of Sets, Subset, Superset, Equivalence Set, Power Set, Union and Intersection
8	Difference, Symmetric Difference, Complement and Properties of Sets
9	Mid-term Exam
10	Ordered Pair, Cartesian Product and Some Properties of Them
11	Relation, Inverse Relation, Composition of Relation, Domain, Range, Equivalence Relations, Equivalence Classes and Partition for Sets
12	Partially Ordered Relations, Least Element, Greatest Element, Minimal Elements, Bounded Below, Bounded Above, Infimum, Supremum, Totally and Well-Ordered Relation
13	Report discussion
14	Final Review
15	Exam

8. Assessment Strategy:

Assessment	Weight
Discussion	5 Marks

Report and Seminar	5 Marks
Quiz (2 quizzes, 3 marks each)	6 Marks
Assignment and Participation	4 Marks
Mid-term Examination	20 Marks
Final Examination	60 Marks
Total	**100%**

9. Student Workload Breakdown

Academic Year: 2026-2027		Semester: Fall	
ECTS: 6		6*27 = 162 Total number of credit hours	
Prerequisite: No			
Type	**No. of Activities**	**Hours (Duration) per Activity**	**Total**
Attendance	14	4 hr	4x14=56
Report	1	10 hr	10x1=10
Mid Term Exam	1	6 hr	6x1=6
Seminar	1	9 hr	9x1=9
Homework	6	5 hr	5x6=30
Discussion	14	2 hr	14x2=28
Quiz	3	3 hr	3x3=9
Site Visit	0	0	0
Project	0	10 hr	10x0=0
Final Exam	1	14 hr	14x1=14
Total			162hr

10. Learning Outcome Mapping:

Topics	**Assessment**	**Week**	**CLO(s)**
Introduction to Foundations of Mathematics	Class Participation	2	CLO1, CLO4
Mathematical Logic, Statements and their Types	Discussion	3	CLO1, CLO2
Tautology, Contradiction, Logical Equivalent	Assignment	4	CLO1, CLO2
Open Sentence, Quantifiers (Existential, Universal)	Quiz	5	CLO1, CLO2
Set Theory (Algebra of Sets)	Discussion	6	CLO1, CLO3

Concept of Sets, Subset, Superset, Equivalence Set, Power Set, Union and Intersection	Assignment	7	CLO1, CLO3
Difference, Symmetric Difference, Complement and Properties of Sets	Quiz	8	CLO1, CLO3
Mid-term Exam	Mid-Term Examination	9	CLO1, CLO2, CLO3
Ordered Pair, Cartesian Product and Some Properties of Them	Discussion	10	CLO1, CLO3
Relation, Inverse Relation, Composition of Relation, Domain, Range, Equivalence Relations, Equivalence Classes and Partition for Sets	Assignment	11	CLO1, CLO2, CLO3
Partially Ordered Relations, Least Element, Greatest Element, Minimal Elements, Bounded Below, Bounded Above, Infimum, Supremum, Totally and Well-Ordered Relation	Discussion	12	CLO1, CLO2, CLO3
Report discussion	Report	13	CLO4, CLO5
Final Review	Class Participation	14	CLO1–CLO6
Exam	Final Examination	15	CLO1–CLO6

11. Sample Examination Paper:

Q1// Define each of the following concepts:
1) Logical equivalent 2) Partition for a set. 3) Greatest lower bound element (infimum).

Q2// Give an example for each of the following:
1) Atomic statement. 2) Composition of two relations.
3) Totally ordered set. 4) An equivalence relation which is not partial order.

Q3// Let $A = \{1,2,3,4\}$, $B = \{2,3\}$ and $R = \{(x, y) \in A \times A;\ y \leq x\}$. Find each of the following with respect to the relation R on A:
(1) R (2) Is R an equivalence relation on A? Why?.
(3) Greatest element of A. (4) Minimal element(s) of A.
(5) $R \Delta R^{-1}$. (6) $P(B) \cup P(A - B)$

Note: This course (subject) book is subject to approval by the Department Council (Scientific Committee) and must be signed by the Head of the Department.